

1. For segment 1 with base = 8192 and limit = 1048, what happens when trying to access the logical address (1, 2056)?

1 point

- ☐ The physical address is 1024.
- ☐ The address is valid with a physical address of 1024.
- ☒ The address is invalid, typically causing a segmentation fault.
- ☐ The physical address is 10248.

2. Which of the following is NOT a key feature of a good main memory management system?

1 point

- ☐ High degree of multiprogramming
- ☐ Efficiency in memory allocation and deallocation
- ☐ Reliable memory allocation to processes
- ☒ Slow memory access times

3. Which of the following statements about the base and limit registers is FALSE?

1 point

- ☐ The base register holds the starting address of a process's memory segment.
- ☒ The MMU adds the value in the limit register to the logical address during address translation.
- ☐ The base and limit registers help in enforcing memory protection.
- ☐ The limit register specifies the size of the memory segment allocated to a process.

4. In the context of overlays, what role does the overlay driver play?

1 point

- ☐ It manages fixed-size memory partitions.
- ☒ It swaps overlays in and out of memory as needed.
- ☐ It generates machine code during program execution.
- ☐ It translates logical addresses to physical addresses.

5. Which of the following statements accurately describes the Next Fit dynamic memory allocation scheme?

1 point

- ☒ It allocates memory from the next available hole, starting from the location of the last allocation.
- ☐ It allocates memory from the first available hole, scanning from the beginning each time.
- ☐ It allocates memory from the largest available hole to minimize fragmentation.
- ☐ It allocates memory from the smallest available hole to minimize wasted space.